FACULTY OF ENGINEERING

INBG 417

Exam Assignment 1

12 June 2026

Nthabiseng Maoka

30224926

Submitted in pursuit of the degree

BACHELOR OF ENGINEERING

In

INDUSTRIAL ENGINEERING

North-West University

Potchefstroom Campus

North-West University: Potchefstroom Campu

Table of Contents

Executive summary

This report presents

DigMore Co., an established shovel manufacturer, is extending its range with an all-purpose folding mini shovel for the camping market, and because its existing plant is already committed to current lines and cannot absorb the new demand, a dedicated facility is required; this report applies a systematic engineering design approach to develop that facility's layout and Master Facility Plan (MFP). The design is sized for a projected demand of 400,000 shovels per year, roughly 1,819 per day on a single nine-hour shift across 220 working days (derived in Section 6) — and the product is built from three aluminium parts: a back handle section, a middle handle section, and a shovel-head assembly of a stamped head and hinge connector riveted together, with a 5% scrap allowance carried at tube cutting and sheet stamping. All materials are stored on site, covering two weeks of raw material and four weeks of finished goods; the powder-coating area is fully enclosed to contain fumes; the site provides public-road access with weekday receiving, on-site collection of small orders and loading space for DigMore's own delivery fleet; and about 10% of the floor area is reserved for projected growth over the next five years. Three alternative block layouts were assessed against [VERIFY: evaluation criteria], and the selected option, Alternative [VERIFY], was refined into a final MFP of approximately [VERIFY: total area] m² covering receiving, storage, production, the enclosed coating area, dispatch, offices and personnel facilities, giving management a defendable basis for establishing the new facility.

List of Tables

List of Figures

List of Equation

1. Executive Summary

2. Academic Integrity and AI Declaration

3. Project Background

DigMore Co. is an established South African shovel manufacturer with a broad customer base across the camping and outdoor market. The company has decided to expand its product range with a new all-purpose folding mini shovel. This compact product consists of two aluminium handle sections (back and middle) manufactured from round tube and a shovel-head assembly produced from aluminium sheet.

The projected annual demand for the folding mini shovel is 400 000 units. DigMore Co.'s existing production facility is already fully utilised by current product lines and cannot absorb this additional volume. Management has therefore decided to establish a new dedicated manufacturing facility for the folding mini shovel only.

An Industrial Engineer has been engaged to develop a comprehensive facility design that meets the projected demand while addressing receiving, storage, production, work-in-progress, finished-goods storage, customer collection and dispatch requirements. The design must also incorporate a fully enclosed powder-coating area for safety and reserve approximately 10% of the floor area for future expansion. This report presents the systematic engineering design process followed, including capacity analysis, process requirements, space planning, alternative layout development and the final Master Facility Plan.

4. Design Aim and Objectives

2.1. Aim

Design a dedicated manufacturing facility for DigMore Co. that meets the projected demand, of 400 000 folding mini shovels per year, provides efficient receiving and dispatch operations, ensures safe and logical material flow, and allows for future expansion.

2.2. Objectives

The following design objectives have been established to guide the development of the new facility:

Production and Capacity

To design a manufacturing facility capable of meeting the annual demand of 400 000 folding mini shovels, incorporating the required production capacity, scrap allowance and future expansion

Site and Access

Recommend a site layout that provides safe access from public roads for supplier deliveries, customer collections, employee access and DigMore Co.'s own delivery fleet.

Layout and Material Flow:

Develop a layout that supports a clear, logical flow of materials from receiving through raw-material storage, production processes, work-in-progress staging, assembly, packaging and finished-goods storage to dispatch.

Equipment and Workstations:

Select appropriate machinery, equipment and workstation designs for all required operations, including cutting, threading, stamping, bending, sharpening, serrating, riveting, heat treatment, sandblasting, polishing, powder coating, assembly and packaging.

Storage and Material handling:

Provide sufficient indoor storage for two weeks of raw materials, four weeks of finished goods and appropriate work-in-progress areas between production stages.

Safety and Support Facilities:

Incorporate all safety requirements, including a fully enclosed powder-coating area, and provide suitable offices, amenities, parking, utilities and other support facilities for management, administrative staff and operators.

5. Design Requirements

The following design requirements, categorised under their relevant headings, have been established to guide the successful design of DigMore Co.'s dedicated folding mini shovel manufacturing facility.

Production requirements

The facility shall:

- Produce 400 000 folding mini shovels per year.
- Provide sufficient production capacity for one 9-hour shift per day, five days per week, based on 220 working days per year.
- Account for operator breaks when determining available productive time per shift.
- Include a 5% scrap allowance at the tube cutting and sheet stamping processes only.
- Provide manual packaging workstations, as packaging also serves as the final visual inspection point before shipping.

FIGURE 1:FOLDING MINI SHOVEL CONCEPT DESIGN

Manufacturing Process requirements

The facility shall:

- Accommodate the complete manufacturing process for the handle cutting, internal threading, grip forming, knurling, external threading, hinge connector stamping, hinge connector bending, shovel head stamping, sharpening, serrating, shovel head bending, riveting, heat treatment, sandblasting, polishing, powder coating, assembly and packaging.
- Provide suitable machines, workstations and operator space for each operation.

- Include work-in-progress staging points between operations where required by the production design.
- Provide process flow charts to support the quality management system and to demonstrate the sequence of manufacturing operations.

Site and access requirements

The facility shall:

- Be accessible from public roads for supplier deliveries, customer collections, employee access and delivery trucks.
- Provide sufficient space for raw-material delivery vehicles to enter, offload and exit the site safely.
- Provide sufficient space for finished-goods dispatch vehicles and company delivery trucks.
- Include a designated collection point for small customer orders.
- Provide space on the premises for the truck fleet.
- Include suitable parking, security control and access control for employees, customers, visitors and goods vehicles.
- Allocate approximately 10% of the floor area for future expansion over the next five years.

FIGURE 2:SITE ACCESS AND VEHICLE MOVEMENT LAYOUT

Layout and Flow requirements

The facility shall:

- Provide a logical flow of materials from receiving to raw-material storage, production, work-in-progress staging, assembly, packaging, finished-goods storage and dispatch.
- Provide clear movement paths for operators, materials and handling equipment between departments.
- Reduce unnecessary backtracking between raw materials, work-in-progress, finished goods and dispatch areas.
- Position finished-goods storage close to dispatch to limit unnecessary movement of completed products.
- Position packaging after assembly and before finished-goods storage to support the final visual check before shipping.

Storage and Material Handling

The facility shall:

- Store all raw materials, components, work-in-progress and finished goods inside the building.
- Provide raw-material storage for two weeks of production stock, including aluminium round tubes, aluminium sheets or coils, and bought-in packaging materials.
- Provide finished-goods storage for four weeks of completed folding mini shovels.
- Provide work-in-progress storage between operations where required by the selected production system.
- Include staging areas for materials and components at relevant manufacturing and assembly workstations.
- Use material-handling methods and aisle widths suited to the selected unit loads, storage arrangements and movement volumes.

[Insert Figure 3.4 here: High-level material flow requirement from receiving to dispatch.]

Figure 3.4: High-level material flow requirement from receiving to dispatch

Receiving and Dispatch

The facility shall:

- Include a designated receiving area for supplier deliveries during weekday working hours.
- Provide space for offloading, inspection and transfer of raw materials to storage or production.
- Include a dispatch area for large orders transported by DigMore Co.'s own delivery trucks.
- Provide a separate or clearly controlled collection point for small customer orders.
- Include staging space for completed orders before loading or customer collection.
- Ensure that receiving and dispatch activities do not obstruct production movement or operator pathways.

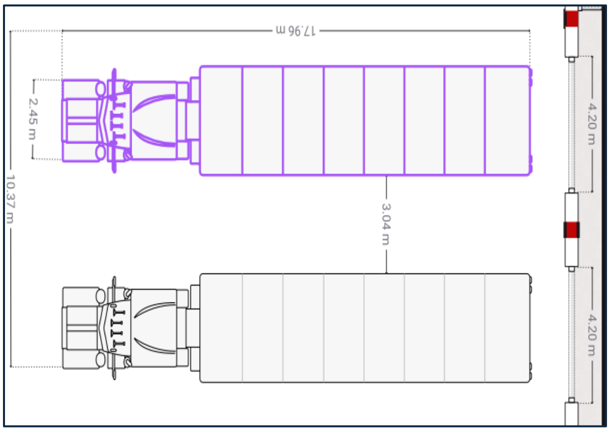

FIGURE 4: LOADING BAY

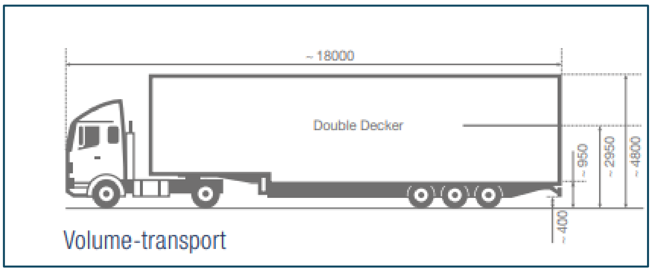

FIGURE 3: TRUCK DIMENSIONS

Safety and Waste Management

The facility shall:

- Locate the powder-coating process in a separate, fully enclosed area within the building due to fume-related safety concerns.
- Provide safe separation where required for cutting, stamping, heat treatment, sandblasting, sharpening, serrating and powder-coating operations.
- Provide clear operator pathways around workstations and material-handling routes.
- Include waste and scrap handling areas, especially for scrap generated during tube cutting and sheet stamping.
- Position scrap and waste collection points so that waste can be removed without disrupting production flow.

Supporting Facilities

The facility shall:

- Provide office space for the Senior Plant Manager, Operations Manager, Inventory Manager, administrative staff, reception, purchasing, sales, finance, marketing, human resources and plant supervisors.

- Provide personnel facilities, including restrooms, lockers and cafeteria access for all employees.
- Provide parking for management, administrative staff, security personnel, operators, customers and company vehicles.
- Include utility requirements such as water, electricity, plumbing, lighting and ventilation.
- Include security facilities for controlling the movement of people, materials and vehicles on site.

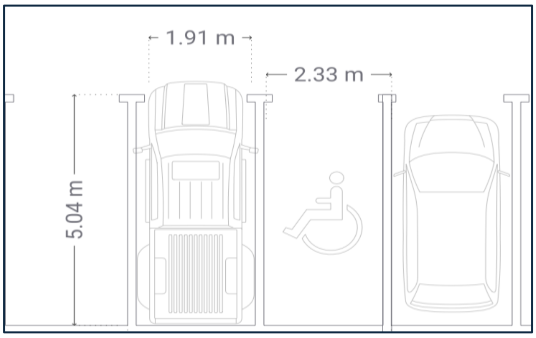

FIGURE 5: PARKING SPACES

Key Layout Implications for Development

The requirements above give rise to the following key implications that will be used when developing and evaluating alternative block layouts:

- Raw-material storage should be positioned close to the first production operations.
- Work-in-progress staging areas should be positioned between related operations without creating excess inventory.
- The enclosed powder-coating area should be separated from general production movement, while still allowing practical entry and exit of parts.
- Packaging must be located after assembly and before finished-goods storage.
- Finished-goods storage should be positioned close to dispatch.
- Receiving, customer collection and dispatch should be arranged so that external vehicle movement does not interfere with internal production flow.
- The 10% future expansion area should be clearly visible on the final Master Facility Plan and should not be used to compensate for current insufficient capacity.

6. Assumptions and Constraints

This section summarises the key assumptions and constraints that underpin the facility design.

6.1. Assumptions

TABLE 1: KEY ASSUMPTIONS USED IN THE FACILITY DESIGN

Assumption	Details
Annual demand	The facility is designed to produve 400 000 folding mini shovels per year.
Operating schedule	One 9-hour shift per day, five days per week, for 220 working days per year.

Productive time per shift	Operators receive one 30-minute lunch break and two 15-minute tea breaks. This gives 480 minutes of productive time per shift before allowances for set-up, maintenance or cleaning
Overtime	Overtime is available only as a flexible capacity measure and is not included in the base design.
Raw-material inventory	Raw-material storage is sized for two weeks of production requirements.
Finished-goods inventory	Finished-goods storage is sized for four weeks of completed shovels
Work-in-progress inventory	WIP storage quantities and locations are determined by the chosen production system and are provided only where staging is required between operations.

6.2. Constraints

The final facility design must satisfy the following constraints:

- The facility must be dedicated to the folding mini shovel product line.
- A 5% scrap allowance applies only to the tube cutting and sheet stamping processes.
- All raw materials, components, work-in-progress and finished goods must be stored inside the facility.
- The powder coating process must be in a separate, fully enclosed area within the building.
- All supplier deliveries, customer collections and finished-goods dispatch activities must occur during weekday working hours.
- Adequate space must be reserved on the premises for DigMore Co.'s own delivery truck fleet.
- Approximately 10% of the facility floor area must be reserved for future expansion within the next five years.

7. Process Flow

The facility layout is developed based on the manufacturing process sequence of the folding mini shovel, ensuring that materials move in a structured and continuous manner through the system.

The process begins with the receiving and storage of raw materials, including aluminium tubes and sheets. These materials are then introduced into the production process, where initial cutting and forming operations are performed.

Subsequently, handle components undergo threading and knurling operations, while sheet metal components are stamped and shaped to form shovel heads. The shovel head components are further processed through sharpening, serrating, and bending operations before being joined to hinge connectors.

Following fabrication, components undergo heat treatment and surface preparation processes, including sandblasting and polishing. Powder coating is then applied to provide the required surface finish, after which final assembly is completed.

The finished product is packaged and transferred to finished goods storage, from where it is either collected by customers or dispatched via company transport.

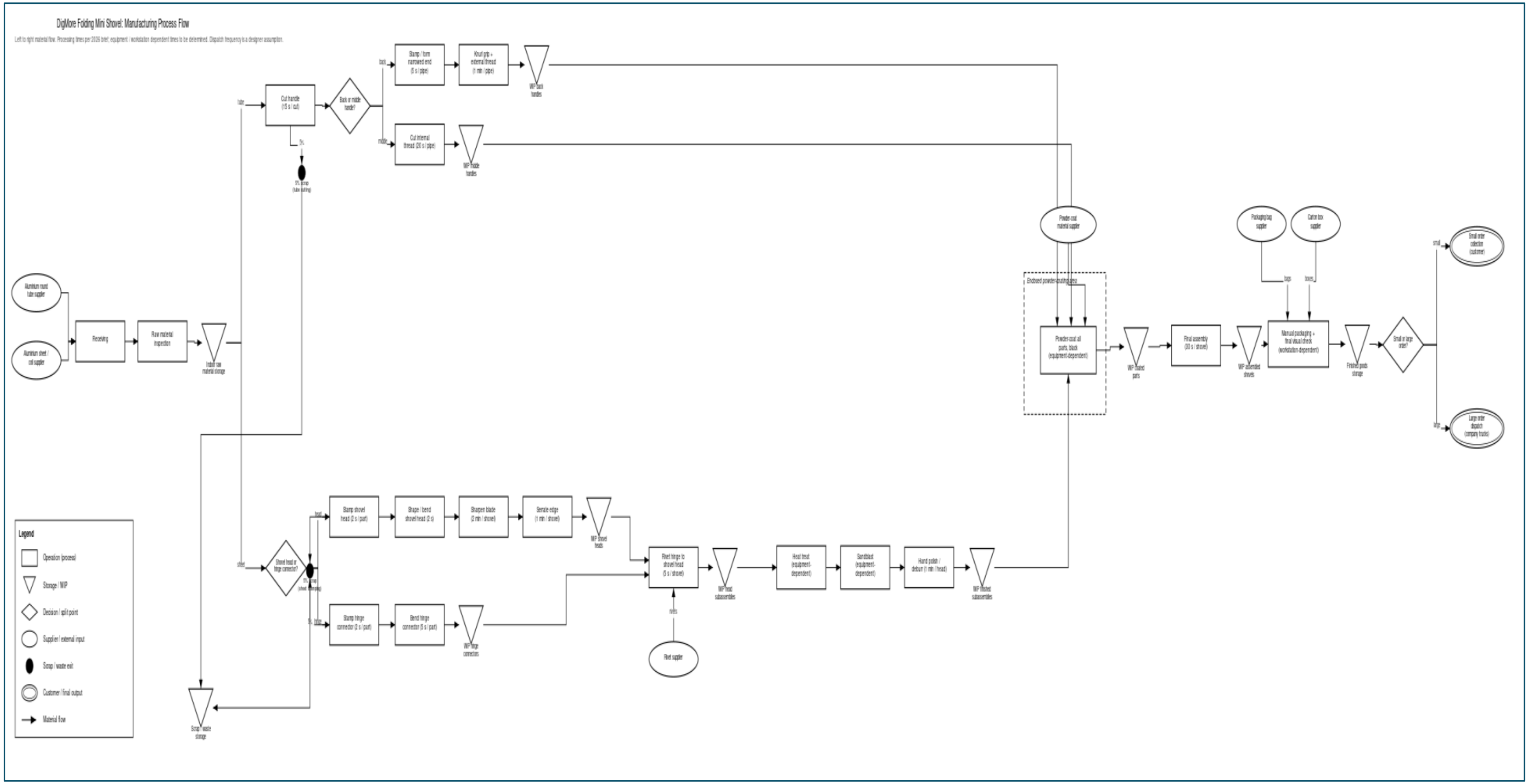

FIGURE 2: PROCESS FLOW FOR FOLDING MINI SHOVEL MANUFACTURING

This process flow forms the foundation for workstation definition, departmental grouping, and the development of alternative facility layouts.

8. Detailed Design Specifications

This section presents the detailed design specifications used to develop the proposed manufacturing facility for DigMore Co. The purpose of this section is to translate the design brief into measurable production, raw material, process, and capacity requirements. The calculations in this section form the basis for later workstation planning, space allocation, department sizing, and alternative layout development.

8.1. Demand

DEMAND

8.2. Raw Material Requirements

6.1.1 Raw material Overview

The raw materials supplied by the supplier are delivered in standard stock forms, primarily in batches of 6 m lengths for tubular materials and sheet/coil form for flat materials. The current supplier for both material types is Aluminium Trading,

- **Aluminium round tubes** (31.70 mm outer diameter × 3.18 mm wall thickness × 6000 mm) are used to manufacture the shovel handle sections (back and middle sections).
- **Rolled aluminium sheets/coils** (2.5 mm thickness) are used to manufacture the shovel head and hinge connector through stamping and forming processes.

	Processing Time
Cut handle	15 s per cut (number of pipes per cut depends on equipment)
Internal thread on the middle section	20 s per pipe
Stamp form end of grip section (narrowing)	5 s per pipe
Knurling and external thread on grip section	1 min per pipe
Stamp hinge connector	2 s per part
Bend hinge connector	5 s per part
Stamp shovel head	2 s per part
Sharpen shovel head blade	2 min per shovel
Serrate shovel head blade edge	1 min per shovel

Shape/bend shovel head	2 s per shovel head
Rivet hinge connector to shovel head	5 s per shovel
Heat treat shovel head subassembly	Depends on equipment
Sandblasting shovel heads	Depends on equipment
Hand polish to remove burrs	1 min per shovel head
Powder-coat all parts (black colour)	Depends on equipment
Assemble shovel (handle + head)	30 s per shovel
Packaging (bagging and boxing)	Depends on workstation design

Table 6.2: Raw Material Item, Dimensions and Use

Raw material	**Dimensions / specification**	**Use**
Aluminium round tube	31.70 mm outside diameter, 3.18 mm wall thickness	Back handle and middle handle sections
Aluminium sheet	2.5 mm thickness	Shovel head and hinge connector
Rivets	Assumed standard aluminium/steel rivets	Joining hinge connector to shovel head
Packaging bags and boxes	Assumed bought-in packaging material	Final packaging

The tube and sheet specifications are taken from the assignment brief. Rivet and packaging details are not fully specified in the brief and are therefore treated as design assumptions.

6.1.3 Finished Goods Demand

The required annual demand is 400 000 folding mini shovels per year. The company operates for 220 workdays per year.

The daily production demand is calculated from the annual demand and the number of working days per year, as shown in Equation 6.1.

Equation 6.1: Daily production demand

$$D_{day} = 400\ 000 / 220$$

$$D_{day} = 1\ 818.18 \approx 1\ 819 \text{ shovels/day}$$

The productive working time per day is calculated from the 9-hour shift less a 30-minute lunch break and two 15-minute tea breaks. This gives 8 productive hours per day.

Using 8 productive hours per day, the hourly production demand is determined as shown in Equation 6.2.

Equation 6.2: Hourly production demand

$$D_{hour} = 1\ 818.18 / 8$$
$$D_{hour} = 227.27 \approx 228 \text{ shovels/hour}$$

TABLE 6.3: FINISHED GOODS DEMAND REQUIREMENTS

Product	Annual demand	Working days/year	Daily demand	Hourly demand
Folding mini shovel	400 000	220	1 819	228

The table converts the annual market demand into daily and hourly production requirements using the operating calendar provided in the brief.

6.1.4 Component Demand

Each finished shovel requires one back handle section, one middle handle section, one shovel head, and one hinge connector. Therefore, the daily component demand before scrap is 1 819 units per component.

Each finished shovel requires one back handle section and one middle handle section. Therefore, the total daily tube-component demand before scrap is calculated as shown in Equation 6.3. A 5% scrap allowance is then applied to determine the required input quantity.

Equation 6.3: Daily tube-component demand before scrap

$$Q_{tube,before} = 1\ 819 + 1\ 819$$
$$Q_{tube,before} = 3\ 638 \text{ components/day}$$

Equation 6.4: Daily tube-component demand after scrap

$$Q_{tube,after} = 3\ 638 / (1 - 0.05)$$
$$Q_{tube,after} = 3\ 829.47 \approx 3\ 830 \text{ components/day}$$

For the stamped sheet-metal components, the required input quantity is calculated using the same scrap-adjustment relationship, as shown in Equations 6.5 and 6.6.

Equation 6.5: Daily shovel-head demand after scrap

$$Q_{head,after} = 1\ 819 / (1 - 0.05)$$
$$Q_{head,after} = 1\ 914.74 \approx 1\ 915 \text{ heads/day}$$

Equation 6.6: Daily hinge-connector demand after scrap

$$Q_{hinge,after} = 1\ 819 / (1 - 0.05)$$
$$Q_{hinge,after} = 1\ 914.74 \approx 1\ 915 \text{ connectors/day}$$

TABLE 6.4: DAILY COMPONENT DEMAND

Component	Components per shovel	Daily demand before scrap	Daily demand after scrap
Back handle section	1	1 819	Included in tube total

Middle handle section	1	1 819	Included in tube total
Total tube components	2	3 638	3 830
Shovel head	1	1 819	1 915
Hinge connector	1	1 819	1 915

The scrap rate is applied only to the processes specified in the assignment brief.

6.1.5 Raw Material Inventory Requirement

The company's raw material inventory policy requires two weeks' worth of raw materials to be stored due to long supplier lead times. Since the company operates five days per week, this corresponds to 10 working days of raw material stock.

The raw-material inventory policy requires two weeks of stock. Since the facility operates five days per week, this corresponds to 10 working days of inventory. The corresponding stock levels are calculated in Equations 6.7 to 6.9.

Equation 6.7: Two-week tube-component stock

$$I_{tube,2wk} = 3\ 830 \times 10$$
$$I_{tube,2wk} = 38\ 300 \text{ tube components}$$

Equation 6.8: Two-week shovel-head stock

$$I_{head,2wk} = 1\ 915 \times 10$$
$$I_{head,2wk} = 19\ 150 \text{ blanks}$$

Equation 6.9: Two-week hinge-connector stock

$$I_{hinge,2wk} = 1\ 915 \times 10$$
$$I_{hinge,2wk} = 19\ 150 \text{ blanks}$$

For tube storage, each finished shovel requires two tube sections of 150 mm each. The required tube length per shovel and the corresponding daily tube length are calculated in Equations 6.10 and 6.11.

Equation 6.10: Tube length required per shovel

$$L_{tube,shovel} = 150 \text{ mm} + 150 \text{ mm}$$
$$L_{tube,shovel} = 300 \text{ mm} = 0.30 \text{ m}$$

Equation 6.11: Daily tube length requirement

$$L_{tube,day} = 1\ 819 \times 0.30$$
$$L_{tube,day} = 545.7 \text{ m/day}$$

Assuming that aluminium tubes are supplied in 6 m stock lengths, the number of stock tubes required per day and for a two-week period is calculated in Equations 6.12 to 6.14.

Equation 6.12: 6 m tubes required per day before scrap

$$N_{tube,before} = 545.7 / 6$$
$$N_{tube,before} = 90.95 \approx 91 \text{ tubes/day}$$

Equation 6.13: 6 m tubes required per day after scrap

$$N_{tube,after} = 90.95 / (1 - 0.05)$$
$$N_{tube,after} = 95.74 \approx 96 \text{ tubes/day}$$

Equation 6.14: Two-week 6 m tube stock

$$I_{tube\ stock,2wk} = 96 \times 10$$
$$I_{tube\ stock,2wk} = 960 \text{ tubes}$$

TABLE 6.5: RAW MATERIAL INVENTORY REQUIREMENT

Raw material / component	Daily requirement after scrap	Two-week stock requirement
6 m aluminium round tubes	96 tubes/day	960 tubes
Shovel head blanks	1 915/day	19 150 blanks
Hinge connector blanks	1 915/day	19 150 blanks

The tube calculation assumes 6 m supply lengths. The assignment brief specifies the tube diameter and handle-section length, but does not explicitly state the supplied tube length; therefore, the 6 m tube length is treated as a design assumption.

8.3. Manufacturing Time and Capacity

The available productive time per day is based on a 9-hour shift less a 30-minute lunch break and two 15-minute tea breaks, as shown in Equation 6.15.

Equation 6.15: Available production time per day

$$H = 8 \times 60$$
$$H = 480 \text{ min/day}$$

Following the textbook machine-fraction method, the machine or workstation requirement is determined from the total daily processing time divided by the available production time per workstation per day.

The number of required machines or workstations is determined by dividing the total daily processing time by the available production time per workstation per day, as shown in Equation 6.16.

Equation 6.16: Required machines or stations

$$N = T_{day} / H$$

The daily processing time required for any operation is obtained from the product of the daily quantity and the processing time per unit, as shown in Equation 6.17.

Equation 6.17: Daily processing time required

$$T_{day} = Q \times t$$

TABLE 6.6: PRODUCTION PROCESSING TIME CONSTRAINTS

Process	Processing time
Cut handle	15 s per cut
Internal thread on middle section	20 s per pipe
Stamp form end of grip section to narrow	5 s per pipe
Knurling and external thread on grip section	1 min per pipe

Stamp hinge connector	2 s per part
Bend hinge connector	5 s per part
Stamp shovel head	2 s per part
Sharpen shovel head blade	2 min per shovel
Serrate shovel head blade edge	1 min per shovel
Shape/bend shovel head	2 s per shovel head
Rivet hinge connector to shovel head	5 s per shovel
Heat treat shovel head subassembly	Depends on equipment
Sandblasting shovel heads	Depends on equipment
Hand polish to remove burrs	1 min per shovel head
Powder-coat all parts	Depends on equipment
Assemble shovel head and handle section	30 s per shovel
Packaging	Depends on workstation design

The processing times are taken from the assignment brief.

8.4. Manufacturing Processes and Workstations

Tube Processing

The Tube Processing Department produces the two aluminium handle components required for each folding mini shovel: the **middle handle section** and the **back handle section**. The middle handle section is cut to length and internally threaded. The back handle section is cut to length, narrowed at one end, externally threaded, and knurled at the grip end. The DigMore brief specifies aluminium round tube with an outside diameter of **31.70 mm** and wall thickness of **3.18 mm**, with a **5% scrap allowance** for tube cutting.

FIGURE 6.5: TUBE HANDLE COMPONENT BREAKDOWN

Insert labelled Draw.io figure here.

Suggested figure content:

BACK HANDLE SECTION
150 mm aluminium tube
Knurled grip section, tube body, and narrowed threaded end.

MIDDLE HANDLE SECTION
150 mm aluminium tube
Internally threaded end and tube body.

FIGURE 6.5: TUBE HANDLE COMPONENT BREAKDOWN

Process Flow

The tube process follows a split-flow arrangement after cutting. Both handle sections begin as cut tube blanks, but the two parts then follow different processing routes before meeting again at prepared-handle WIP storage.

Raw Tube Staging
↓
Tube Cutting
↓
Component Split

↓ ↓
Middle Handle Back Handle
↓ ↓
Internal Threading End Narrowing
↓ ↓
↓ External Threading
↓ ↓
↓ Knurling

↓
Prepared Handle WIP
↓
Final Assembly

FIGURE 6.6: TUBE PROCESSING PROCESS FLOW

Insert Draw.io version of the process flow above.

Machine

The selected equipment below is provisional and should be treated as the first design selection for workstation sizing. Each process is linked to a selected machine, a machine specification table, a reference figure, and a workstation layout, following the same structure for every department in this section.

TABLE 6.9: TUBE PROCESSING EQUIPMENT SELECTION

Operation	Selected equipment	Reason for selection	Source
Tube cutting	HIPPO CS425CNC-O high-speed aluminium cutting machine	Suitable for aluminium and soft-metal cutting; listed cutting capacity includes round pipe up to Ø150 mm × 6 mm; machine size is 2200 × 1100 × 2200 mm.	hippobender.com
Internal threading	SFX-M30R / M6–M30 electric tapping machine	Suitable for aluminium tapping; tapping range includes M6–M30; machine size is listed as 810 × 770 × 220 mm.	brokentapremover.com
End narrowing	SG40NC pipe end forming / shrinking machine	Suitable for tube shrinking/reducing; listed maximum tube capacity is Ø40 × 2 mm and machine	liyemachine.en.made-in-china.com

		size is 2000 × 550 × 1300 mm.	
External threading	Asada B50-AT pipe threading machine	Listed pipe capacity is 1/2″–2″; automatic die-head version with 750 W motor.	cltoolcentre.com.au
Knurling	Round tube embossing / knurling machine	Listed as a round tube embossing/knurling machine with size 1400 × 600 × 1500 mm.	alibaba.com
Tube WIP cart	FlexQube tube cart	Listed as a tube cart for transporting long tubes, with dimensions 3080 × 630 mm and load capacity of 2425 lb.	flexqube.com
Scrap bin	Industrial steel waste / scrap bin	Local South African suppliers manufacture heavy-duty steel waste containers for scrap metal and industrial offcuts; exact size should be requested from supplier.	macnay.co.za

FIGURE 6.7: TUBE CUTTING MACHINE REFERENCE
INSERT IMAGE OF HIPPO CS425CNC-O TUBE CUTTING MACHINE HERE.

FIGURE 6.8: INTERNAL THREADING / TAPPING MACHINE REFERENCE
INSERT IMAGE OF SFX-M30R OR SIMILAR ELECTRIC TAPPING MACHINE HERE.

FIGURE 6.9: END-FORMING / NARROWING MACHINE REFERENCE
INSERT IMAGE OF SG40NC PIPE END FORMING MACHINE HERE.

FIGURE 6.10: EXTERNAL THREADING MACHINE REFERENCE
INSERT IMAGE OF ASADA B50-AT PIPE THREADING MACHINE HERE.

FIGURE 6.11: KNURLING MACHINE REFERENCE
INSERT IMAGE OF ROUND TUBE KNURLING/EMBOSSING MACHINE HERE.

FIGURE 6.12: TUBE PROCESSING WORKSTATION LAYOUT
INSERT DRAW.IO LAYOUT HERE.

FIGURE 6.13: TUBE CART / RAW TUBE HANDLING REFERENCE
INSERT IMAGE OF TUBE CART OR LONG-STOCK TROLLEY HERE.

Machines

The following tables record provisional machine specifications for layout sizing and capacity justification. Final selection must still be confirmed with suppliers.

TABLE 6.10: TUBE CUTTING MACHINE SPECIFICATIONS

Equipment specification	**Value**
Selected machine	HIPPO CS425CNC-O high-speed aluminium cutting machine

Main function	Cutting aluminium and other soft-metal tube/profile material
Cutting capacity	Round pipe up to Ø150 mm × 6 mm
Cutting accuracy	±0.05 mm
Machine dimensions	2200 × 1100 × 2200 mm
Operator requirement	1 operator assumed
Cost	Supplier quotation required
Source	hippobender.com

Note: The DigMore tube wall thickness is 3.18 mm, while the SG40NC source lists Ø40 × 2 mm. This machine is useful as a reference for layout sizing, but the final selected machine must be confirmed with a supplier for **31.70 mm OD × 3.18 mm wall aluminium tube**.

Table 6.13: External Threading Machine Specifications

Equipment specification	**Value**
Selected machine	Asada B50-AT pipe threading machine
Main function	External pipe/thread cutting
Pipe capacity	1/2″–2″
Spindle speed	40 rpm
Motor power	750 W
Machine dimensions	640 × 570 × 500 mm
Machine weight	71 kg
Operator requirement	1 operator assumed
Cost	Supplier quotation required
Source	generaltools.com.au

Table 6.14: Knurling Machine Specifications

Equipment specification	**Value**
Selected machine	Round tube embossing / knurling machine
Main function	Tube grip pattern forming
Machine dimensions	1400 × 600 × 1500 mm
Machine weight	330 kg
Power	1500 W
Required air pressure	50 litres listed by supplier

Operator requirement	1 operator assumed
Cost	Supplier quotation required
Source	alibaba.com

Tube Processing Workstation

The workstation requirement is based on the Section 6.2 capacity analysis and the selected provisional equipment.

Table 6.15: Tube Processing Workstation Requirements

Operation	**Processing time**	**Selected equipment**	**Stations required**
Tube cutting	15 s per cut	HIPPO CS425CNC-O cutting machine	2
Internal threading	20 s per pipe	SFX-M30R electric tapping machine	2
End narrowing	5 s per pipe	SG40NC end forming machine	1
External threading and knurling	1 min per pipe	Asada B50-AT + round tube knurling machine	4

Processing times are from the DigMore brief.

Tube Processing Workstation

FIGURE 6.14: TUBE PROCESSING WORKSTATION LAYOUT

Legend:
[RM] Raw material
[M] Machine
[W] WIP storage
[S] Scrap/offcut bin
[OA] Operator aisle

[OA] Operator Aisle

[RM] Raw Tube Rack
6 m tube stock

[M] Cutting x 2

[S] Scrap Bin

↓

[W] Cut Blank Sorting

Middle Handle Line
[M] Internal Tapping x 2
[W] Middle Handle WIP

Back Handle Line
[M] End Narrowing x 1
[M] External Thread x 4
[M] Knurling x 4

[W] Back Handle WIP

[W] Prepared Handle WIP to Assembly

Material Handling

Material handling follows a consistent logic across the facility: raw material is staged, processed parts are transferred using carts, WIP-before and WIP-after are clearly shown, and scrap is collected at the process where it is generated.

TABLE 6.16: TUBE PROCESSING MATERIAL HANDLING

Flow	Handling method	Destination	Figure placeholder
Raw tube to cutting	Tube rack / tube cart	Tube cutting station	Figure 6.13
Cut blanks to sorting	Small WIP bin/cart	Component split point	Figure 6.14
Middle handle blanks	WIP bin/cart	Internal threading station	Figure 6.14
Back handle blanks	WIP bin/cart	End narrowing station	Figure 6.14
Finished handle parts	Labelled WIP cart	Final assembly	Figure 6.15
Tube scrap/offcuts	Steel scrap bin	Waste handling area	Figure 6.16

FIGURE 6.14: CUT BLANK WIP BIN / SORTING CART
INSERT IMAGE OF SMALL WIP BIN OR TROLLEY HERE.

FIGURE 6.15: PREPARED HANDLE WIP CART
INSERT IMAGE OF LABELLED WIP TROLLEY HERE.

FIGURE 6.16: SCRAP / OFFCUT BIN
INSERT IMAGE OF INDUSTRIAL STEEL SCRAP BIN HERE.

Tube Processing Area Calculation

The final area must be refined after exact supplier equipment is selected. The following calculation gives a defensible first-pass layout allowance.

Equation 6.3: Tube Processing Area

$$A_{tube} = (A_{machines} + A_{raw} + A_{WIP} + A_{scrap} + A_{aisle}) \times 1.10$$

Table 6.17: Machine Footprint Calculation

Equipment	Quantity	Footprint per unit	Total footprint
Tube cutting machine	2	2.2 × 1.1 = 2.42 m²	4.84 m²
Internal tapping machine	2	0.81 × 0.77 = 0.62 m²	1.24 m²

End-forming machine	1	2.0 × 0.55 = 1.10 m²	1.10 m²
External threading machine	4	0.64 × 0.57 = 0.36 m²	1.44 m²
Knurling machine	4	1.4 × 0.6 = 0.84 m²	3.36 m²
Total machine footprint			**11.98 m²**

Machine dimensions used in this table are taken from the selected equipment sources.

Table 6.18: Tube Processing Area Allowance

Area component	**Value**	**Basis**
Machine footprint	11.98 m²	From Table 6.17
Raw tube staging	6.00 m²	Assumed 6 m × 1 m raw tube rack space
WIP sorting and prepared handle WIP	4.00 m²	Assumed WIP bins/carts for split flow
Scrap/offcut bin	3.25 m²	Assumed 2.5 m × 1.3 m scrap bin footprint
Operator and trolley movement allowance	26.00 m²	Design allowance for safe movement between machines
Subtotal	**51.23 m²**	
10% expansion allowance	5.12 m²	DigMore brief requires about 10% expansion space
Preliminary tube processing area	**56.35 m²**	
Recommended rounded area	**57 m²**	Use in Section 6.5 space summary

The Tube Processing Department should therefore be allocated a preliminary area of approximately **57 m²**. This area includes the selected machine footprints, raw tube staging, WIP storage, scrap/offcut collection, operator movement, trolley movement, and the 10% future expansion allowance.

TABLE 6.19: TUBE PROCESSING DEPARTMENT SUMMARY

Item	**Final value / decision**
Department input	Aluminium round tube
Department output	Prepared back and middle handle sections
Main equipment	Cutting, threading, end-forming, knurling
Main bottleneck risk	External threading and knurling
Scrap source	Tube cutting
Material handling	Tube rack, WIP carts, scrap bin

Next department	Final assembly
Preliminary department area	**57 m²**

Sheet and Hinge Processing

The Sheet and Hinge Processing department stamps and forms the shovel head and hinge connector from 2.5 mm aluminium sheet, in parallel with tube processing, before both streams converge at riveting.

Sheet-Metal Press Cell

Equipment class: Gap-frame mechanical press with dies

Market example: Stamtec OCP series gap-frame press

Machines / operators: 2 presses / 2 operators

Cell size: 6.4 m x 4.0 m

WIP before: Sheet packs or coil-fed staging

WIP after: Head blank bins and hinge blank bins

Scrap handling: Skeleton scrap bin

Safety requirement: Guards or light curtains; die-change access

[Figure pending: gap-frame mechanical press for sheet stamping; reference stamtec.com]

Hinge Bending and Shovel Head Shaping

Hinge bending: small electric press brake, 1 station, placed after hinge stamping.

Shovel head shaping: forming press / bending machine, 1 station, placed after head stamping and before sharpening.

[Figure pending: electric press brake for hinge bending; reference cotswold-machinery-sales.co.uk]

Finishing Operations

Finishing covers blade sharpening, serrating, hand polishing and riveting. Sharpening is the major workload driver in the facility and should be treated as a bottleneck department fed by a controlled WIP lane.

Blade Sharpening Bank

Equipment class: Belt grinder with extraction

Market example: Fein GRIT GI75 abrasive belt grinder

Machines / operators: 8 machines / 8 operators

Bank size: 8.4 m x 6.2 m

WIP before: Central WIP lane

WIP after: Trolley or bin transfer to serrating

Safety requirement: Eye protection, hearing protection, dust extraction, lockout for belt changes

[Figure pending: abrasive belt grinder for sharpening; reference echopkins.com]

Blade Serrating and Hand Polishing

Blade serrating: profiled abrasive / serrating stations, 4 stations, in-line after sharpening.

Hand polishing / deburring: manual benches with extraction, 4 stations, before coating route.

[Figure pending: serrating station reference; cotswold-machinery-sales.co.uk]

Riveting Workstation

Equipment class: Pneumatic radial riveting machine

Market example: AGME radial riveting machine

Machines / operators: 1 machine / 1 operator

Cell size: 3.7 m x 2.6 m

WIP before: Matched head and hinge connector buffers

WIP after: Head subassembly bin or trolley

Safety requirement: Two-hand control, task lighting, compressed-air drop

[Figure pending: radial riveting machine; reference agme.net]

Heat Treatment and Surface Preparation

Heat treatment and sandblasting prepare the riveted head subassembly for powder coating. Both are equipment-dependent per the brief and must be finalised after oven and booth selection.

Heat Treatment Area

Equipment class: Aluminium ageing batch oven

Market example: Wisconsin Oven aluminium age oven

Area allowance: 80 m2 zone including oven and buffers (matches Section 6.6 heat treat and sandblast allowance)

WIP basis: Day buffer before and after heat treatment

Safety requirement: Thermal clearance, hot-part handling, PPE, ventilation

[Figure pending: aluminium ageing batch oven; reference wisoven.com]

Sandblasting Area

Equipment class: Enclosed blast booth or blast room

Market example: Airblast AFC blast rooms

WIP before / after: Racked subassemblies

Safety requirement: Dust collection, PPE, entry control, sealed waste drums

[Figure pending: enclosed sandblasting booth; reference airblastafc.com]

Powder Coating

Powder coating is located in a separate, fully enclosed area due to fume-related safety concerns, per the design constraints. The enclosure coats the handle sections and the head subassembly before final assembly.

Equipment class: Conveyorised powder-coating line

Market example: P.I. Marketing powder-coating system

Layout elements: Load/hang station, booth, cure oven, cooling leg, unload station, air lock

Area allowance: 135 m2 (from Section 6.6 space summary)

Safety requirement: Fully enclosed area, extraction, make-up air, controlled entry/exit

[Figure pending: conveyorised powder-coating line; reference pimarketing.co.za]

Final Assembly and Packaging

Final assembly is where the handle stream and coated head subassembly converge. Packaging is performed manually and also serves as the final visual check before finished goods are transferred to storage, per the design requirements.

Final assembly equipment: manual benches with holding fixtures, 2 stations / 2 operators.

Packaging equipment: manual packing benches, bag/carton storage, pallet-build position, 2 stations / 2 operators.

Quality control: final visual check and reject/quarantine bin.

Handling method: hand pallet truck to finished-goods storage.

[Figure pending: final assembly bench / hand pallet truck; reference uline.com, adendorff.co.za]

Workstation footprints above are indicative for equipment layout only. The aggregate production floor allowance used for overall building sizing is the capacity-workbook figure in Section 6.6 (238 m2 for 34 machines), pending a full workstation-by-workstation layout reconciliation.

8.5. Receiving and Dispatch

Receiving and Dispatch

Storage and Material Handling
Supporting Facilities
Space Requirements

Space Requirements

The space requirements below are derived from the capacity and demand model (single source of truth workbook), applying the production, storage, personnel and site assumptions set out in the Assumptions and Constraints section. Figures marked as a designer assumption remain to be verified against supplier or civil quotations before final sign-off.

Production Floor Space

Area / department	Floor area (m2)	Basis
Production workstations	238	34 machines x 7 m2 per workstation (designer assumption, incl. operator space)
Powder coating (enclosed)	135	Supplier-based estimate; conveyorised line, fume extraction, air lock
Heat treat and sandblast	70	Supplier-based estimate; T6 ageing oven and sandblast booth

Storage Space Requirements

Raw material and finished goods quantities are drawn from the demand and inventory model (2-week raw material cover, 4-week finished goods cover). Rack and stacking arrangements are designer assumptions to be verified once racking suppliers are engaged.

Store	Items	Racks / positions	Floor area (m2)	Basis
Raw tube	900 tubes	5	37.5	Cantilever racking, 200 tubes per rack (designer assumption)
Raw sheet	230 sheets	2	9.4	Sheet stacks, 150 sheets per stack (designer assumption)
Finished goods	102 pallets	26	46.8	Palletised, racked, 4 storage levels (designer assumption)

WIP staging	6 units	2	3.6	Decoupling buffers, approximately one shift
Scrap / waste	-	-	30	Designer assumption

Total storage area: 127.3 m2.

Supporting Facilities Space Requirements

Personnel numbers combine the named roles required by the brief with designer-assumed support staff and the 34 production operators derived 1:1 from the capacity model.

Category / role	**Number**	**Note**
Senior Plant Manager	1	Brief
Operations Manager	1	Brief
Inventory Manager	1	Brief
Reception	1	Brief
Purchasing	1	Brief
Sales	2	Brief (assumed count)
Finance	1	Brief
Marketing	1	Brief
Human Resources	1	Brief
Plant supervisors	3	Designer assumption, approximately one per department
Security	2	Designer assumption
Material handlers / forklift operators	4	Designer assumption
Equipment-dependent and QA operators	5	Heat treat, sandblast, powder coat, inspection (designer assumption)
Production operators	34	From capacity model, line processes 1:1
TOTAL headcount	58	

Office and amenity space:

Area / department	**Floor area (m2)**	**Basis**

Receiving and dispatch	220	Docks and staging (designer assumption)
Offices	216	24 non-operator staff x 9 m2 per person (designer assumption)
Amenities (lockers, restrooms, cafeteria)	203	58 headcount x 3.5 m2 per person (designer assumption)

Parking and site vehicle provision:

Item	**Number / area**	**Basis**
Employee bays	58 bays	1 bay per on-site employee
Visitor and collection bays	8 bays	Designer assumption
Total car bays	66 bays, 1 848 m2	28 m2 per bay including aisle (designer assumption)
Truck fleet bays	5 bays, 350 m2	Own fleet, 70 m2 per bay including manoeuvring (designer assumption)
Total vehicle area	2 198 m2	

Total Building and Site Footprint

Item	**Floor area (m2)**
Subtotal (functional floor area)	1 209.3
Circulation and aisle allowance (25%)	302.3
Building subtotal	1 511.6
Future expansion provision (approximately 10%)	151
TOTAL BUILDING FOOTPRINT	1 663
Parking and truck fleet (site)	2 198
INDICATIVE SITE FOOTPRINT	3 861

All figures above are traceable to the capacity and space workbook and will be refined once supplier quotations are obtained for enclosed process areas, racking and paving rates.

9. Design Alternatives

Three alternative block layouts were developed from the activity relationships implied by the process flow (Section 5) and the department groupings used in the space requirements above

(Section 6.6). Each alternative arranges the same set of departments (receiving, raw material storage, tube processing, sheet and hinge processing, heat treat and surface preparation, the enclosed powder coating area, assembly and packaging, finished goods storage, dispatch, and support facilities) into a different flow pattern, so that the choice between them turns on flow logic, hazard separation and expansion capacity rather than on department content.

Alternative 1: Linear Spine

A single-direction flow from receiving to dispatch, with departments laid out left to right along one production spine. This gives the most direct, easiest-to-follow material flow and the simplest construction sequence. Its main weakness is that the enclosed powder-coating area sits mid-line, which lengthens the fume-extraction duct run and brings coating traffic closer to pedestrian routes than is ideal.

Alternative 2: U-Shape Return

Receiving and dispatch share one face of the building, with the process flow wrapping around in a U so that both external-facing activities are on the same side. This is the most compact option and suits efficient internal material handling, but it crowds the powder-coating enclosure against active production areas and leaves the least room for the required 10% expansion margin within a tight footprint.

Alternative 3: Parallel-Branch with Finishing Spine

Tube processing and sheet/hinge processing run as two parallel halls, each feeding into a perpendicular finishing, assembly and packaging spine. Powder coating sits as a separated block off the spine with its own dedicated extraction, giving the strongest hazard separation of the three options and a clear direction (an additional feeder cell) in which to apply the 10% expansion allowance. The trade-off is a somewhat higher build complexity and more pedestrian crossings where the two feeder halls meet the spine.

Evaluation Criteria and Weighting

Each alternative was scored from 1 (poor) to 5 (excellent) against seven criteria drawn directly from the design requirements and constraints, weighted by their relative importance to the brief. The weighted score for each alternative is the sum of its criterion scores multiplied by the criterion weight.

Criterion	Weight
Meets demand of 400 000 shovels/yr	0.20
Logical material flow (receiving to dispatch)	0.20
Enclosed powder coating and hazard separation	0.15
Efficient storage and material handling	0.15
Pedestrian / vehicle safety and site access	0.10
Approximately 10% expansion provision	0.10
Ease of construction / implementation	0.10

Weighted Evaluation

Criterion	Weight	Alt 1: Linear Spine	Alt 2: U-Shape	Alt 3: Parallel-Branch
Meets demand of 400 000/yr	0.20	5	5	5
Logical material flow	0.20	5	3	4
Enclosed powder coating / hazard separation	0.15	3	2	5
Efficient storage and material handling	0.15	3	4	3
Pedestrian / vehicle safety	0.10	4	3	4
Approximately 10% expansion provision	0.10	3	2	5
Ease of construction	0.10	5	4	3
Weighted total	**1.00**	**4.10**	**3.40**	**4.20**

Alternative 3 (Parallel-Branch with Finishing Spine) scores highest overall, driven by its stronger hazard separation and expansion margin, both of which are explicit brief requirements (fully enclosed powder coating; approximately 10% expansion). Alternative 1 remains competitive on flow simplicity and ease of construction, while Alternative 2's compactness is offset by its weaker expansion margin. Alternative 3 is therefore carried forward as the basis for the Master Facility Plan.

All figures in this section are derived from the capacity and space workbook (single source of truth). The scoring reflects a first-pass designer judgement against the stated criteria and should be revisited once the Master Facility Plan is drawn in detail.

[Figure pending: block layout sketches for Alternatives 1, 2 and 3 - see Alternatives.drawio.pdf for prior exploratory layouts, to be finalised against the department areas in Section 6.6]